



Calcite

△ **Calcite crystals** displaying a vivid purple colouring

Although calcite forms spectacular crystals of varied shapes and in virtually all colours, most calcite occurs in the form of limestone, marble, or travertine, all of which are used as ornamental and carving stones. Travertine is a dense, banded rock formed by the evaporation of river and spring waters, depositing coloured layers of calcite. Sliced travertine and marble were used extensively as a facing stone for buildings in ancient Greece and Rome, and many ancient Egyptian “alabaster” carvings are actually calcite.

Specification

Chemical name Calcium carbonate | **Formula** CaCO_3
Colours Colourless, white, various | **Structure** Trigonal
Hardness 3 | **SG** 2.7 | **RI** 1.48–1.66 | **Lustre**
Vitreous | **Streak** White | **Locations** Iceland, USA,
Germany, Czech Republic, Mexico



Calcite and apatite | Rough | Calcite commonly occurs with other gemstones, usually providing a later infilling, as with these apatite crystals.



Stunning crystals | Rough | Calcite has the largest number of different crystal structures of any mineral, including these perfectly formed scalenohedrons.



Scalenohedron | Rough | Scalenohedrons, a form of calcite, are essentially high-angle, hexagonal pyramids, as seen in this single, well-defined example.



Faceted stone | Cut | This superb stone from Tanzania was faceted by a master cutter. Calcite is remarkably difficult to facet because it is soft and breaks easily.

The Viking sunstone

Crystal compass

Seafaring Vikings relied on the sun for navigation, but cloudy days posed a problem. Norse sagas talk of a “sunstone” that could help find its position on any day, grey or bright. Some scientists think this may have been calcite: a calcite crystal polarizes light into two beams, which can be lined up to find the location of the sun. Modern calcite “detectors” achieve this within 1 per cent accuracy.



Viking hunters This rock painting at Alta in Norway depicts two Vikings fishing – one casts a net. They may have been guided by a sunstone.

Calcite alabaster | Carved | Ancient Egyptians used calcite for items such as this canopic stopper from Tutankhamun's tomb.



Eyes picked out in black paint

Headdress of carved calcite